MSG No. 57.387 $P_{b}bcm$ [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0,\ \frac{3}{4},\ 0\right]$	`[1,2,13,14]`
2	$\left[0,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[0,\ \frac{1}{4},\ 0\right]$	`[5,6,9,10]`
4	$\left[0,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[7,8,11,12]`

Table 2: Wyckoff site: `4b`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{3}{4},\ 0\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{4},\ 0\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[7,8,11,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `2'/m'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,5,10,14]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,6,9,13]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,8,11,15]`

Table 4: Wyckoff site: `4d`, site symmetry: `2'/m'..`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,5,10,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,6,9,13]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,8,11,15]`

Table 5: Wyckoff site: `4e`, site symmetry: `m'2'm`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,8,11,14]`
2	$\left[0,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[2,7,12,13]`
3	$\left[0,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[3,6,9,16]`
4	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[4,5,10,15]`

Table 6: Wyckoff site: `4f`, site symmetry: `m'2'm`

No.	position	mapping
1	$\left[\frac{1}{2}, y, \frac{1}{4}\right]$	`[1,8,11,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{3}{4}\right]$	`[2,7,12,13]`
3	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{4}\right]$	`[3,6,9,16]`
4	$\left[\frac{1}{2}, -y, \frac{3}{4}\right]$	`[4,5,10,15]`

Table 7: Wyckoff site: `8g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, \frac{3}{4}, 0\right]$	`[1,2]`
2	$\left[-x, \frac{1}{4}, \frac{1}{2}\right]$	`[3,4]`
3	$\left[-x, \frac{1}{4}, 0\right]$	`[5,6]`
4	$\left[x, \frac{3}{4}, \frac{1}{2}\right]$	`[7,8]`
5	$\left[x, \frac{1}{4}, 0\right]$	`[9,10]`
6	$\left[-x, \frac{3}{4}, \frac{1}{2}\right]$	`[11,12]`
7	$\left[-x, \frac{3}{4}, 0\right]$	`[13,14]`
8	$\left[x, \frac{1}{4}, \frac{1}{2}\right]$	`[15,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `2'..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,10]`
2	$\left[x, \frac{1}{2}, 0\right]$	`[2,9]`
3	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[3,12]`
4	$\left[-x, 0, \frac{1}{2}\right]$	`[4,11]`
5	$[-x, 0, 0]$	`[5,14]`
6	$\left[-x, \frac{1}{2}, 0\right]$	`[6,13]`
7	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[7,16]`
8	$\left[x, 0, \frac{1}{2}\right]$	`[8,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `m'..`

No.	position	mapping
1	$[0, y, z]$	`[1,14]`
2	$\left[0, \frac{1}{2}-y, -z\right]$	`[2,13]`
3	$\left[0, y+\frac{1}{2}, \frac{1}{2}-z\right]$	`[3,16]`
4	$\left[0, -y, z+\frac{1}{2}\right]$	`[4,15]`
5	$[0, -y, -z]$	`[5,10]`
6	$\left[0, y+\frac{1}{2}, z\right]$	`[6,9]`
7	$\left[0, \frac{1}{2}-y, z+\frac{1}{2}\right]$	`[7,12]`
8	$\left[0, y, \frac{1}{2}-z\right]$	`[8,11]`

Table 10: Wyckoff site: 8j, site symmetry: m’..

No.	position	mapping
1	$\left[\frac{1}{2}, y, z\right]$	[1,14]
2	$\left[\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[2,13]
3	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[3,16]
4	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	[4,15]
5	$\left[\frac{1}{2}, -y, -z\right]$	[5,10]
6	$\left[\frac{1}{2}, y+\frac{1}{2}, z\right]$	[6,9]
7	$\left[\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[7,12]
8	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	[8,11]

Table 11: Wyckoff site: 8k, site symmetry: ..m

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	[1,8]
2	$\left[x, \frac{1}{2}-y, \frac{3}{4}\right]$	[2,7]
3	$\left[-x, y+\frac{1}{2}, \frac{1}{4}\right]$	[3,6]
4	$\left[-x, -y, \frac{3}{4}\right]$	[4,5]
5	$\left[x, y+\frac{1}{2}, \frac{1}{4}\right]$	[9,16]
6	$\left[x, -y, \frac{3}{4}\right]$	[10,15]
7	$\left[-x, y, \frac{1}{4}\right]$	[11,14]
8	$\left[-x, \frac{1}{2}-y, \frac{3}{4}\right]$	[12,13]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[x, \frac{1}{2}-y, -z\right]$	[2]
3	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[3]
4	$\left[-x, -y, z+\frac{1}{2}\right]$	[4]
5	$[-x, -y, -z]$	[5]
6	$\left[-x, y+\frac{1}{2}, z\right]$	[6]
7	$\left[x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[7]
8	$\left[x, y, \frac{1}{2}-z\right]$	[8]
9	$\left[x, y+\frac{1}{2}, z\right]$	[9]
10	$[x, -y, -z]$	[10]
11	$\left[-x, y, \frac{1}{2}-z\right]$	[11]
12	$\left[-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[12]
13	$\left[-x, \frac{1}{2}-y, -z\right]$	[13]
14	$[-x, y, z]$	[14]
15	$\left[x, -y, z+\frac{1}{2}\right]$	[15]
16	$\left[x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[16]